

work that we do is being able to interface those models into any of the use cases that you have. So the translation work that happens, whether it's quantization or taking it and translating into a particular use case that we want. So the way we work with these ISP's is making that translation. So what necessarily happens right now is utilising the heterogeneous architecture more effectively. So take the model, run it effectively. We decide what's the right place to run it, be it the CPU, GPU or NPU and then we modify it. So that's the work that you'll see significant that will happen between now and launch.

Alex: And memory optimisation. So you want models to actually reside on the device and the beauty of it is if you don't have any connection to the cloud, you could still run all your Gen AI stuff on the device. And then if you're connected, what happens is you use a hybrid situation. So, you run some things on the cloud where you run on the processing power here and then the cloud. Also look at it the other way. The cloud will offload to the device so that cost doesn't run up in your data center to a point where it's not manageable. Because if the user interface changes and people start using these added functions just like everyone's expecting to do then the queries going back to the cloud would be so high that you run out of bandwidth and you run out of performance capability. It will take longer time to come back. You want some privacy, you want personalisation, you want immediacy. So you run things on the device. We started with a 1 billion parameter Stable Diffusion Gen AI model. We ran it on the phone. Now you can draw what you want to see. It generates it, but we ran it in 20 seconds before and now it's running on one second. Less than one second and then you super res (upscale) it to a resolution that you want. Now we have 7 billion parameters models running on the handset PCs running 30 billion parameters. So we can optimise these things down to the memory that's contained within the device and then start running better and better AI functions.

Digit: So are you collaborating with platforms like Hugging Face on which a lot of people submit their own models and then you'll have your official variants as well?

Alex: We're collecting models for use cases on our solutions that people can pick and choose from. They can also go to Hugging Face and get those same models from there that have been optimised to our solution. Hugging face, Llama, Baichuan in China, we work with every one. We're kind of agnostic towards that. Plus also run times such as ONNX, PyTorch, Gemini, etc, we work with all of them. And what we do is kind of route them to our API interfaces that go directly down to our NPU for best acceleration. So we've tested so many



models and we're going to continue doing those and make them available for our customers.

Digit: So is there some Git repositories where developers can go and pick these optimized models. Or do they have to reach out to Qualcomm separately?

Alex: Both.

Digit: Regarding developers, there needs to be a significant effort involved in getting more and more developers to sort of build for ARM because that's a challenge. And if you don't have native apps then you have to rely on compatibility layers. And those things sometimes it works and sometimes it doesn't. So, what kind of efforts are you taking to help developers out?

Kedar: Almost two years back, we, with Microsoft and Satya, announced this development platform, called Volterra,

using a Snapdragon based platform that we've been handing out to developers for many couple years now. Think of it as a three-step approach, one you want the app to run on the device. That means we need, even if it's emulated, we make sure that with Microsoft the emulator performance is excellent, so at least it works flawlessly. Second step is to port the app and be native and so we've already got a large set of applications that we've already publicly talked about. We can get you the list of all these applications that have been ported. Things like Amazon Prime, all the the top hundred 300 apps that you see, a lot of them are already native. And then the third step is where a lot of these apps take advantage of the different courses because that's where you'll start to see the benefit of the architecture and. Folks like Adobe we've talked about, they they were on stage with us. You have people like Amazon. There are many, many ISVs, hundreds of ISVs, people like camo, people like Beethoven, etc. So lots of these ISVs that have actually gone to step #3 and optimised. So we've got hundreds of ISVs now running on our NPU. We have several other initiatives like this. We have our own device cloud setup where we have all of these devices in where ISVs or even developers can remotely log in. We give them full access to SDKs, APIs, all of the documentation, etc. So we feel pretty good about where we are. And from an ecosystem perspective, we've been doing a lot of partnership events to make sure that the ecosystem is ready at launch.

Alex: Microsoft's been super supportive of us as well, and, you know, making sure backwards compatibility is there. The emulation platform is actually running better than another emulation running on another device. It is actually better than that. So I think we're in a pretty good place. Of course, the work will continue with bringing on more and more ISVs and making sure that they could take advantage of what we have making those available to our OEM's and partners to make that happen. And just continue to work that way. 